Physics I

Code: EBA1110

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Course Content

- Lecture 1: Units and Vectors.
- Lecture 2: Newton Laws of motion &Gravitational Force, weight and mass
- Lecture 3: Work, Kinetic and Potential Energy, Work Energy theorem& Conservation law of Energy

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- Lecture 4: Oscillatory motion , simple and Harmonic Motion, Introduction to mechanical waves properties , and Travelling wave
- Lecture 5: Sound wave Power, Intensity, speed in solid , liquid or gas , Sound level calculation
- Lecture 6: Superposition of waves.

Course Content

- Lecture 7: capacitors.
- Lecture 8: Calculation of the Electric Current, Resistance, Ohm's law, Power Calculation.
- Lecture 9: Electromotive force , Internal Resistance , Resistance in Series and Parallel, Kirchhoff law and RC Circuit
- Lecture 10: Magnetic Force
- Lecture 11: Biot Savart Law, Ampere Law and their applications. Magnetic force for two parallel conductors.Explanation of Faraday Law of Induction

LECTURE 2

- **Force and Newton Laws of motion**
- **Gravitational Force, weight and mass**

1 -The Concept of Force

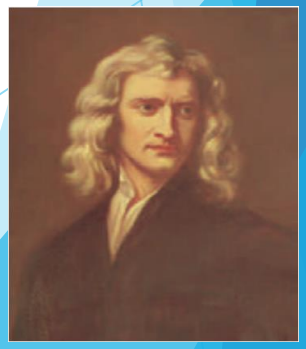
Everyone has a basic understanding of the concept of force from everyday experience. When you push your empty dinner plate away, you exert a force on it. Similarly, you exert a force on a ball when you throw or kick it.

force refers to:
an interaction with an object by means of muscular activity and some change in the object's velocity

Forces do not always cause motion, however. For example, when you are sitting, a gravitational force acts on your body and yet you remain stationary

What force causes the Moon to orbit the Earth?

Newton answered this and related questions by stating that forces are what cause any change in the velocity of an object. The Moon's velocity changes in direction as it moves in a nearly circular orbit around the Earth. This change in velocity is caused by the gravitational force exerted by the Earth on the Moon.



classes of force

contact forces

they involve physical contact between two objects

Examples

- When a coiled spring is pulled
- When a stationary cart is pulled

. When a football is kicked

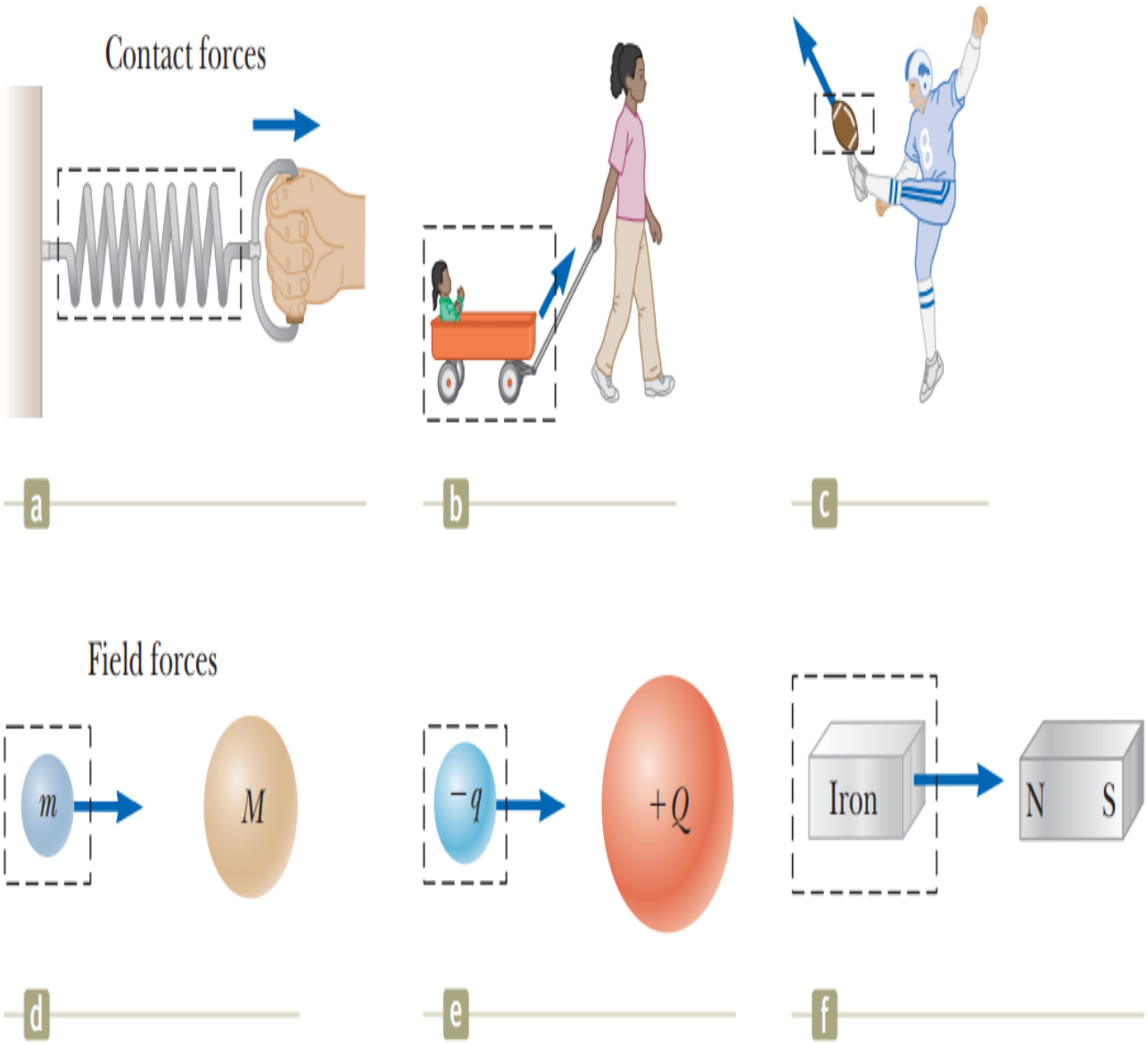
field forces

does not involve physical contact between two objects. These forces act through empty space

Examples

- The gravitational force of attraction between two objects with mass
- the electric force Between electric charges
- the force a bar magnet exerts on a piece of iron

Figure 5.1 Some examples of applied forces. In each case, a force is exerted on the object within the boxed area. Some agent in the environment external to the boxed area exerts a force on the object.



The Vector Nature of Force

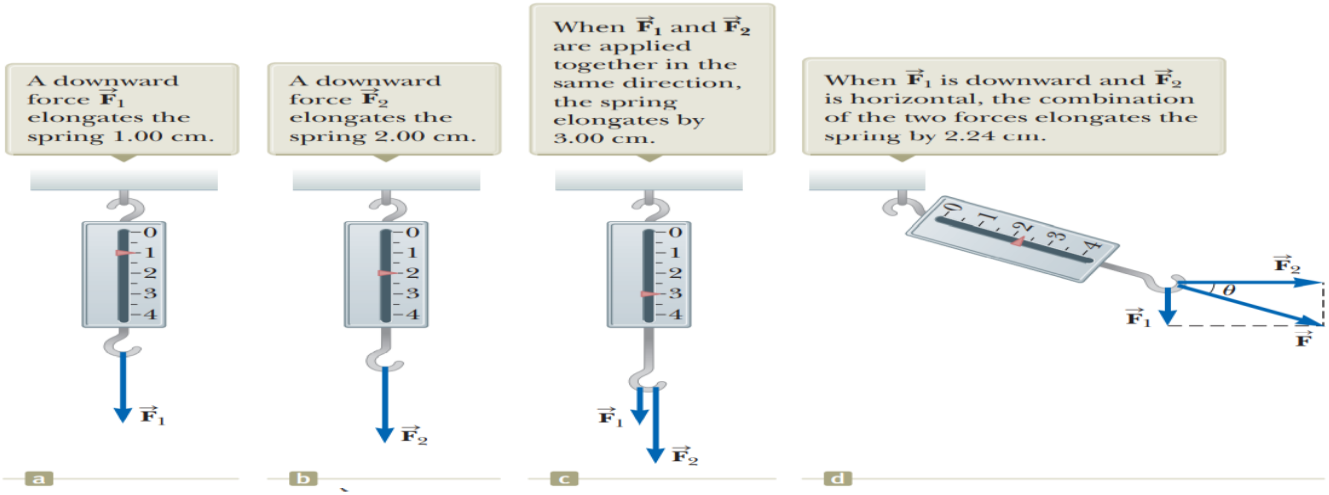


Figure 5.2 The vector nature of a force is tested with a spring scale

It is possible to use the deformation of a spring to measure force. Suppose a vertical force is applied to a spring scale that has a fixed upper end as shown in Figure 5.2a. The spring elongates when the force is applied, and a pointer on the scale reads the extension of the spring. We can calibrate the spring by defining a reference force $\vec{\mathbf{F}}_1$ as the force that produces a pointer reading of 1.00 cm. If we now apply a different downward force $\vec{\mathbf{F}}_2$ whose magnitude is twice that of the reference force $\vec{\mathbf{F}}_1$ as seen in Figure 5.2b, the pointer moves to 2.00 cm. Figure 5.2c shows that the combined effect of the two collinear forces is the sum of the effects of the individual forces.

Now suppose the two forces are applied simultaneously with $\vec{\mathbf{F}}_1$ downward and $\vec{\mathbf{F}}_2$ horizontal as illustrated in Figure 5.2d. In this case, the pointer reads 2.24 cm. The single force $\vec{\mathbf{F}}$ that would produce this same reading is the sum of the two vectors $\vec{\mathbf{F}}_1$ and $\vec{\mathbf{F}}_2$ as described in Figure 5.2d. That is, $|\vec{\mathbf{F}}_1| = \sqrt{F_1^2 + F_2^2} = 2.24$ units, and its direction is $\theta = \tan^{-1}(-0.500) = -26.6°$. Because forces have been experimentally verified to behave as vectors, you *must* use the rules of vector addition to obtain the net force on an object.

2 -Newton's First Law and Inertial Frames

Newton's first law of motion, sometimes called the law of inertia, defines a special set of reference frames called inertial frames. This law can be stated as follows:

If an object does not interact with other objects, it is possible to identify a reference frame in which the object has zero acceleration

Such a reference frame is called an inertial frame of reference.

Another statement of Newton's first law:

In the absence of external forces and when viewed from an inertial reference frame, an object at rest remains at rest and an object in motion continues in motion with a constant velocity (that is, with a constant speed in a straight line).

In other words, when no force acts on an object, the acceleration of the object is zero. From the first law, we conclude that any isolated object (one that does not interact with its environment) is either at rest or moving with constant velocity. The tendency of an object to resist any attempt to change its velocity is called inertia.

from the first law, we can define force as:
that which causes a change in motion of an object

Mass and weight:

Mass is that property of an object that specifies how much resistance an object exhibits to changes in its velocity

Mass is an inherent property of an object and is independent of the object's surroundings and of the method used to measure it. Also, mass is a scalar quantity and thus obeys the rules of ordinary arithmetic

The weight of an object is equal to the magnitude of the gravitational force exerted on the object and varies with location

3-Newton's second law:

When viewed from an inertial reference frame, the acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass:

$$\vec{\mathbf{a}} \propto \frac{\sum \vec{\mathbf{F}}}{m}$$

If we choose a proportionality constant of 1, we can relate mass, acceleration, and force through the following mathematical statement of Newton's second law:[1]

$$\sum \vec{\mathbf{F}} = m\vec{\mathbf{a}}$$

the *net force* $\sum \vec{\mathbf{F}}$ is the vector sum of all forces acting on the object

(We sometimes refer to the net force as the *total force,* the *resultant force,* or the *unbalanced force.*)

it is imperative to determine the correct net force on an object.

Many forces may be acting on an object, there is only one acceleration.

Equation 2 is a vector expression and hence is equivalent to three component equations:

$$\sum F_x = ma_x \qquad \sum F_y = ma_y \qquad \sum F_z = ma_z \qquad (3)$$

The SI unit of force is the **newton** (N). A force of 1 N is the force that, when acting on an object of mass 1 kg, produces an acceleration of 1 m/s^2. From this definition and Newton's second law, we see that the newton can be expressed in terms of the following fundamental units of mass, length, and time:

$$1\ \text{N} \equiv 1\ \text{kg} \cdot \text{m/s}^2$$

In the U.S. customary system, the unit of force is the **pound** (lb). A force of 1 lb is the force that, when acting on a 1-slug mass,[2] produces an acceleration of 1 ft/s^2:

$$1\ \text{lb} \equiv 1\ \text{slug} \cdot \text{ft/s}^2$$

A convenient approximation is 1 N $\approx \frac{1}{4}$ lb.

5- The Gravitational Force and Weight

All objects are attracted to the Earth. The attractive force exerted by the Earth on an object is called the **gravitational force** $\vec{\mathbf{F}}_g$. This force is directed toward the center of the Earth,[3] and its magnitude is called the **weight** of the object.

We saw in Section 2.6 that a freely falling object experiences an acceleration $\vec{\mathbf{g}}$ acting toward the center of the Earth. Applying Newton's second law $\sum \vec{\mathbf{F}} = m\vec{\mathbf{a}}$ to a freely falling object of mass m, with $\vec{\mathbf{a}} = \vec{\mathbf{g}}$ and $\sum \vec{\mathbf{F}} = \vec{\mathbf{F}}_g$, gives

$$\vec{\mathbf{F}}_g = m\vec{\mathbf{g}} \tag{5}$$

Therefore, the weight of an object, being defined as the magnitude of $\vec{\mathbf{F}}_g$, is given by

$$F_g = mg \tag{6}$$

Because it depends on g, weight varies with geographic location. Because g decreases with increasing distance from the center of the Earth, objects weigh less at higher altitudes than at sea level.

6- Newton's Third Law

If two objects interact, the force $\vec{\mathbf{F}}_{12}$ exerted by object 1 on object 2 is equal in magnitude and opposite in direction to the force $\vec{\mathbf{F}}_{21}$ exerted by object 2 on object 1:

$$\vec{\mathbf{F}}_{12} = -\vec{\mathbf{F}}_{21} \qquad (5.7)$$

example

A force F is applied to an object of mass m_1 produces an acceleration of 3.00 m/s^2. The same force is applied to a second object of mass m_2 produces an acceleration of 1.00 m/s^2.

(a) What is the value of the ratio m_1/m_2?

(b) If $m1$ and $m2$ are combined into one object, find its acceleration under the action of the force F.

solution

For the same force F, acting on different masses

$$F = m_1 a_1$$

and

$$F = m_2 a_2$$

(a) $$\frac{m_1}{m_2} = \frac{a_2}{a_1} = \boxed{\frac{1}{3}}$$

(b) $$F = (m_1 + m_2)a = 4m_1 a = m_1(3.00\ \text{m/s}^2)$$

$$a = \boxed{0.750\ \text{m/s}^2}$$

5.7 Analysis Models Using Newton's Second Law

In this section, we discuss two analysis models for solving problems in which objects are either in equilibrium ($\vec{\mathbf{a}} = 0$) or accelerating under the action of constant external forces. Remember that when Newton's laws are applied to an object, we are interested only in external forces that act on the object. If the objects are modeled as particles, we need not worry about rotational motion. For now, we also neglect the effects of friction in those problems involving motion, which is equivalent to stating that the surfaces are *frictionless*. (The friction force is discussed in Section 5.8.)

We usually neglect the mass of any ropes, strings, or cables involved. In this approximation, the magnitude of the force exerted by any element of the rope on the adjacent element is the same for all elements along the rope. In problem statements, the synonymous terms *light* and *of negligible mass* are used to indicate that a mass is to be ignored when you work the problems. When a rope attached to an object is pulling on the object, the rope exerts a force on the object in a direction away from the object, parallel to the rope. The magnitude T of that force is called the **tension** in the rope. Because it is the magnitude of a vector quantity, tension is a scalar quantity.

Analysis Model: The Particle in Equilibrium

If the acceleration of an object modeled as a particle is zero, the object is treated with the **particle in equilibrium** model. In this model, the net force on the object is zero:

$$\sum \vec{\mathbf{F}} = 0 \tag{5.8}$$

Analysis Model: The Particle Under a Net Force

If an object experiences an acceleration, its motion can be analyzed with the **particle under a net force** model. The appropriate equation for this model is Newton's second law, Equation 5.2:

$$\sum \vec{\mathbf{F}} = m\vec{\mathbf{a}} \tag{5.2}$$